Chapter 1 – Polynomial Functions

domain/range

- **Domain**
  - All sets of x-values must be real numbers for a function to be polynomial.
- **Range**
  - EVEN- can have a restriction on Y-values since both end behaviors end on the same side.
  - ODD- cannot have a restriction since the polynomial must pass through the x-axis, following end behavior

factor expressions, factor equations to solve

graph a polynomial function as taught in class

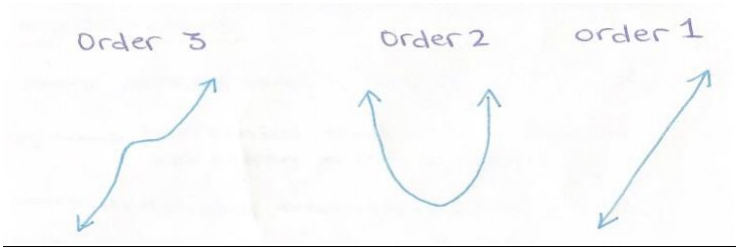
THIS INFO HELPS YOUR GRAPH A POLYNOMIAL

EVEN POLYNOMIALS

- **End behavior**
  - The END BEHAVIOUR tells you where the arrows of a polynomial will end. For an even polynomial, they always end in the same direction.
  - POSITIVE LEADING COEFFICIENT = Q2 →Q1
  - NEGATIVE LEADING COEFFICENT = Q3→Q4
- **Degree**
  - The degree tells you much about the polynomial.
  - If the function is factored, you must add up all the exponents to find the degree. If the degree is an even number, then the polynomial is even.
  - NUMBER OF ZEROS = can be as many as the degree. It can also be zero since the function does not have to cross the x-axis to get to its proper quadrant.
  - TURNING POINTS = One less than the degree. An even polynomial's degrees can only be an odd amount. It must have at least one turn.

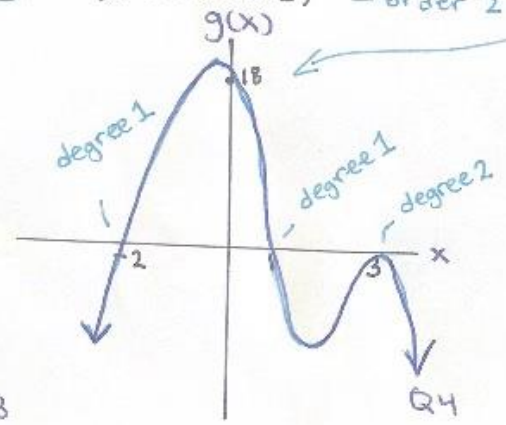
ODD POLYNOMIALS

- **End Behaviour**
  - The end behavior of an Odd polynomial is always in the opposite directions
  - POSITIVE LEADING COEFFICIENT = Q3→Q1
  - NEGATIVE LEADING COEFFICIENT = Q2→Q4
- **Degree**
  - If the degree is odd, then the polynomial is odd
  - NUMBER OF ZEROS = can be as many as the degree, has to at least be one because it must cross the x-axis
  - TURNING POINTS = one less than the degree. An odd polynomial must have only an even amount of turns. It does not need to have any turns at all.



$g(x) = -(x+2)^1(x-1)^1(x-3)^2$ — order 2

flipped



y - int = 18

$f(0.1) = 15.9$

$f(-0.1) = 20$

Maxes out on left side

Degree = 4

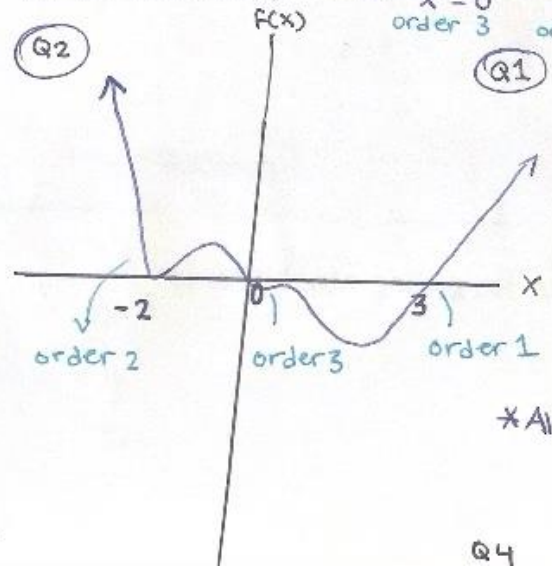
# of TP = 1, 3

End Behaviour = Q3 - Q4

# of zeros = 0, 1, 2, 3, 4

This helps you figure out which side order 2 is higer on.

$f(x) = 2x^3(x-3)(x+2)^2 \longrightarrow$ $2x^3 = 0$, $x = 0$, order 3 ; $(x-3)$, $x = 3$, order 1 ; $(x+2)^2$, $x = -2$, order 2



Y = int = 0 → Found by plugging in zero

Degree = 6 → add up the degrees when its not in standard form.

Possible turning pts = 3 - 3 x-values as shown from factored form.

# of zeros = 3 - as shown

End behaviour = Q2 - Q1

since degree is even (6) and positive, end behavior is this.

*Always label graph

transformation of a point

The parent function $f(x) = x^3$ is transformed to $f(x) = 3[-\frac{1}{4}(x+1)]^3 + 5$

State transformations $\qquad f(x) = a(k(x+d)) + c$

$a = 3$ ⟶ vertical stretch by a factor of 3 (|3|)

$k = -\frac{1}{4}(4)$ ⟶ horizontal stretch by a factor of 4 ($\frac{1}{4}$)
Reflection in the y-axis. (-k)

$d = -1$ ⟶ Horizontal translation 1 unit to the left

$c = 5$ ⟶ vertical translation 5 units up.

Where would the point $(\underset{x}{-2}, \underset{y}{-8})$ on the parent function be transformed to?

$\left(\frac{x}{k} + d,\ ay + c\right)$

⟶ Plug in known variables ⟶ $\left(\frac{x}{-1/4} - 1,\ 3y + 5\right)$

Plug in (x, y)

$\frac{-2}{-1/4} - 1,\ 3(-8) + 5$

$= 7, -19$

Chapter 2 – Polynomial Equations and Inequalities

The Factor Theorem

Factor Theorem

$x - a$ **is a factor of** $f(x)$ **if and only if** $f(a) = 0$

The Remainder Theorem

Remainder Theorem

When $f(x)$ **is divided by** $x - a$ **the remainder is** $f(a)$

factor / solve a higher degree polynomial (n = 3 *or* 4) expression or equation

The division statement is..	2. Divisor – what you are dividing by
Dividend = divisor x quotient + remainder	3. Quotient – what you conclude/solve
1. Dividend – what you start with	4. Remainder – what is remaining

Express equation in factored form, must use factor theorm & synthtic ÷

$f(x) = 9x^4 + 6x^3 - 35x^2 - 24x - 4$

① Find Factors of f(x)

$\frac{b}{a}$ = factors a = first number, b = last number { You must find all factors of (a) and (b), then divide them to find factors of f(x).

(a = 9) – ±1, 3, 9
(b = 4) – ±1, 2, 4

$x = \pm 1, \frac{1}{3}, \frac{1}{9}, 2, \frac{2}{3}, \frac{2}{9}, 4, \frac{4}{3}, \frac{4}{9}$

These are all possible x-values

② Use factor theorm to determine if they are x-values of f(x)

f(1) = −48 ∴ Not factor
f(−1) = −12 ∴ Not factor
f(2) = 0 ∴ Yes a factor!
This makes the equation = 0, it is an x-value

③ Use synthetic division to find the rest of the x-values

divisor

2	9	6	−35	−24	−4
	↓	18	48	26	4 +
x	9	24	13	2	0
	x^3	x^2	x^1	x^0	R

- Put the x-value on the left
- Make sure #s are in decending order according to degree. If there is a missing degree, add a zero in place of it.
- Bring down the first term, and multiply it by the divisor
- add the number by the next term, repeat

Write the new equation

$(9x^3 + 24x^2 + 13x + 2)$

Synthetic division reminder: if divison is a value greater than 1 (Ex. (2x−3) then factor out the two and divide by it after synthetic division has been done. DON'T divide the remainder though.

④ Find the rest of the x-values

Since you are unable to factor by grouping, you must repeat process 2–3 again

$f(-\frac{1}{3})$ = is a factor!

$-\frac{1}{3}$	9	24	13	2
	↓	−3	−7	−2 +
x	9	21	6	0
	x^2	x^1	x^0	R

⑤ Use factoring or quadratic formula

$(9x^2 + 21x + 6)$

$\frac{-21 \pm \sqrt{225}}{18} = x = -1/3,\ x = -2$

⑥ Write Conclusion statement

* remember to include x-values found using factor theorm

$x = \{-\frac{1}{3}, -2, 2\}$ $(9x^4 + 6x^3 - 35x^2 - 24x - 4) = (x-2)(x+2)(3x+1)^2$

lowest to highest write original equation first, then write factors. two x-values so, add squared

$x+3$ is a factor of $f(x) = cx^3 + dx^2 + x - 15$.
remainder is 55 when the function is divided by $x-2$.
Solve for c & d.

- we can solve this using substitution

① $x = -3$, it is a factor so, $y = 0$

$0 = c(-3)^3 + d(-3)^2 + (-3) - 15$ - set $y=0$ and plug in (-3) for x.

$0 = -27c + 9d - 18$ - simplify

$\frac{0}{9} = \frac{-27c}{9} + \frac{9d}{9} - \frac{18}{9}$ - divide everything by 9 to simplify

$0 = -3c + d - 2$ > rearrange to isolate d

$d = 3c + 2$

② $y = 55$ when $x = 2$

Do the same as #1 but, don't isolate (d)

$55 = c(2)^3 + d(2)^2 + 2 - 15$

$55 = 8c + 4d - 13 - 55$

$0 = \frac{8c}{4} + \frac{4d}{4} - \frac{68}{4}$

$0 = 2c + d - 17$

③ Plug in equation 1 into equation 2

EQ1 - $d = 3c + 2$

EQ2 - $0 = 2c + d - 17$

equivalet to (d)

$0 = 2c + (3c + 2) - 17$

$0 = 2c + 3c + 2 - 17$ - simplify

$0 = \frac{5c}{5} - \frac{15}{5}$ - solve for c

$c = 3$

④ Plug in (c) into equation for (d)

Since $c = 3$, you can use it to find (d)

$d = 3c + 2$

$d = 3(3) + 2$

$d = 11$

∴ Therefore

$c = 3$

$d = 11$

linear polynomial inequalities

Interval Notation = $[3, \infty]$

Set Notation = $x \geq 3$

$$9x + 4 \geq 2x - 10$$

– make sure x is (+)

$$9x - 2x \geq -10 - 4$$

$$\frac{7x}{7} \geq \frac{-14}{7}$$

$$x \geq -2$$

$$\frac{-2x}{-2} > \frac{10}{-2}$$

$$x < -5$$

Sign flips when dividing by a negative

$$7x + \frac{5}{3} \geq -5x - 13$$

$$\frac{12x}{12} \geq \frac{-\frac{44}{3}}{12}$$

$$x \geq -\frac{11}{9}$$

$$\frac{2x+3}{5} \leq \frac{4x-7}{3}$$

• multiply both sides to remove denom

$3(2x+3) \leq 5(4x-7)$ – simplify

$6x + 9 \leq 20x - 35$ – get x to one side

$6x - 20x \leq -35 - 9$

$\frac{-14x}{-14} \leq \frac{-44}{-14} \longrightarrow x \geq \frac{22}{7}$

flip sign

Solve using factor table $= -2(3x+2)(x-5)^2 \geq 0$

– This is asking, when is the function greater than zero (above x-axis)

① Identify all x-values, factor if needed

$$\boxed{x = -\frac{2}{3},\ 5}$$

• Always add $-\infty$, ∞ at the beginning and end

② create factor table

	$-\infty$	-3	$-2/3$	4	5	6	∞
$-2 = 0$		–		–		–	
$(3x+2)$		–		+		+	
$(x-5)^2$		+		+		+	
Overall		+		–		–	

pick points between the values

all x-value equations go here

tells you if its (+) or (–)

$\therefore x \leq -\frac{2}{3}$

This tells me that the function is greater than 0 after $-\frac{2}{3}$.

factor sum / difference of cubes

- **Difference of cubes**

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

- **Sum of cubes**

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$\underbrace{(x-1)}_{a}{}^3 - \underbrace{(2x+3)}_{b}{}^3$$

$$= (a-b)(a^2+ab+b^2)$$

Expand

$$= \big((x-1)-(2x+3)\big)\big((x-1)^2+(x-1)(2x+3)+(2x+3)^2\big)$$

$(x-1)(x-1)$ $\quad$ $(2x+3)$ $(2x+3)$

$$(x-1-2x+3)(x^2-2x+1+2x^2+x-3+4x^2+12x+9)$$

$$= (-x-4)(7x^2+11x+7)$$

$$= -(x+4)(7x^2+11x+7)$$

$27x^3-64y^3$ → take the cube root of both numbers

$(3x)^3-(4y)^3$ → put it into formula

$$= (a^3-b^3) = (a-b)(a^2+ab+b^2)$$

$$= (3x-4y)((3x)^2+(3x)(4y)+(4y)^2)$$

Simplify $-(3x-4y)(9x^2+12xy+16y^2)$

$$\frac{x^3}{8}-\frac{27y^3}{64}$$

$\frac{1}{64}(8x^3-27y^3)$ - remove fraction

$\frac{1}{64}((2x)^3-(3y)^3)$ - cube root

$\frac{1}{64}(2x-3y)(4x^2+6xy+9y^2)$

↳ simplify

word problem

The temp of the container is $T(m) = 6m^3-25m^2+29m-7$

When is the temp at $\underline{3^\circ}$.

-Do the same steps you would to solve a equation

$3 = 6m^3-25m^2+29m-7$ - set $=0$

$0 = \underline{6m^3}-25m+29m-10$

① Factor Theorm

$f(1)=0$ $(x-1)$

② Synthetic Division

$=6x^2-19x+10$

③ Factor

$\underbrace{6x^2-4x}-\underbrace{5x+10}$

$2x(3x-2)-5(3x-2)$

$(2x-5)(3x-2)$

④ Therefore...

Temp is $\underline{3^\circ}$ at 1 min, $\frac{2}{3}$ min, $\frac{5}{2}$ min

$x = \{\frac{2}{3}, 1, 5/2\}$

Chapter 3 – Rational Functions, Equations and Inequalities

graph a rational function

$$f(x) = \frac{-2x + x^2 - 15}{x^2 + 6 + 5x}$$

Unit 3

Rearrange and factor $- \dfrac{(x-5)\cancel{(x+3)}}{(x+2)\cancel{(x+3)}}$

cancel eachother, this tells me there will be a hole at $x = -3$.

Restrictions

denominator $\neq 0$

If denom is 0, then function isn't true.

x-value restrictions

$x \neq -2$

$x \neq -3$

These are found from the denom.

Domain: $\{x \in \mathbb{R} \mid x \neq -2, -3\}$

Holes: $x = -3$ When you cancel something from numerator & denominator, then there is a hole.

Find hole point

$f(-3) = \dfrac{x-5}{x+2} = 8$

$(-3, 8) =$ hole

Asymptotes –

VA – What make $f(x) = 0$

$x = -2 \quad x = -3$

HA – If denom is larger than numerator $y = 0$

If denom is = to numerator $y =$ Num / denom

If denom is less than numerator, there is a slant asymptote

Same so $y = 1$ ← $\dfrac{x^2 - 2x - 15}{x^2 + 5x + 6}$ $\quad HA = y = 1$

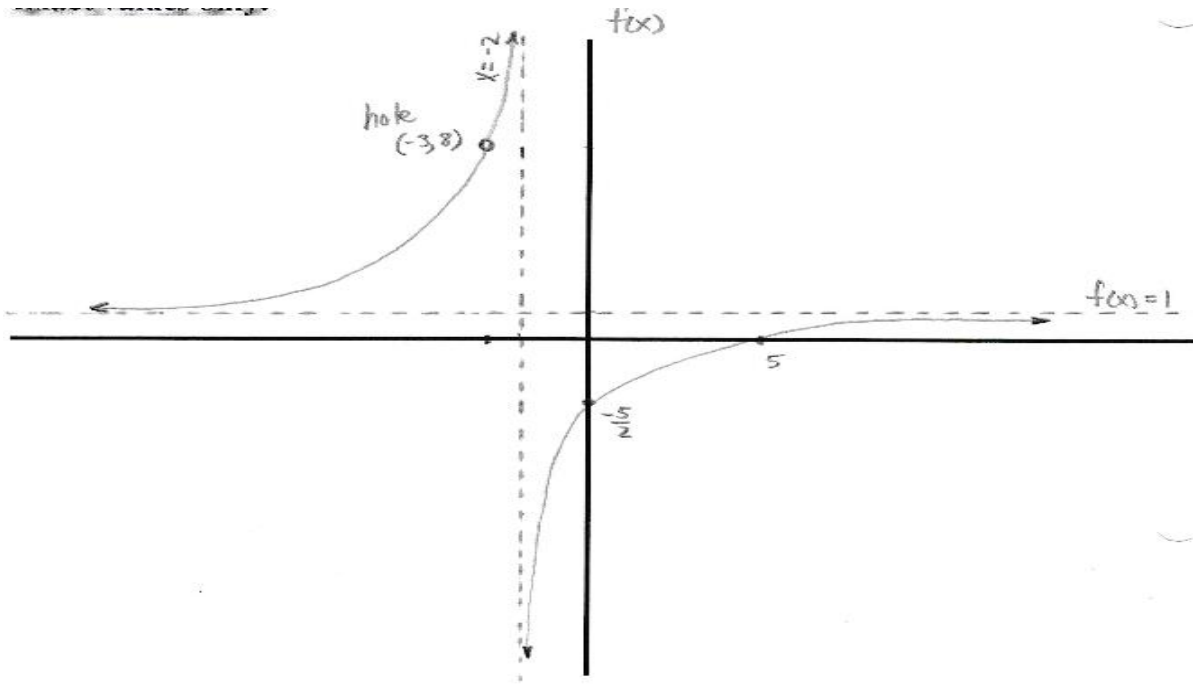
x int – Let $f(x) = 0$ for the numerator

$(0) = x - 5$

$x = 5$ $\quad (5, 0)$ ordered pair

y int – Let $x = 0$ for whole equation

$f(0) = \dfrac{0-5}{0+2} \quad 0 = -\dfrac{5}{2} \quad \left(0, -\dfrac{5}{2}\right)$



Range - $\{ f(x) \in \mathbb{R} \mid f(x) \neq 1, 8 \}$

Positive Interval(s)	Negative Interval(s)
$(-\infty,-3) \cup (-3,-2) \cup (5,\infty)$	$(-2,5)$

Decreasing Interval(s)	Increasing Interval(s)
never	$(-\infty,-3) \cup (-3,-2) \cup (-2,\infty)$

solve rational equation

Solve for x. $\dfrac{x-1}{7x+6} = \dfrac{2}{x+2}$

- Begin by multiplying both sides by lowest common denom to remove fraction.

LCD: $(7x+6)(x+2)$

Restrictions
$x \neq -\frac{6}{7}, -2$

$$\cancel{(7x+6)}\cancel{(x+2)}\frac{x-1}{\cancel{7x+6}} = \cancel{(7x+6)}\cancel{(x+2)}\frac{2}{\cancel{x+2}}$$

$$(x+2)(x-1) = 2(7x+6)$$

Put into standard form

$$x^2 + x - 2 = 14x + 12$$

Set equation = to zero

$$x^2 + x - 2 - 14x + 12 = 0$$

$$x^2 - 13x - 14 = 0$$

Expand and simplify again

$$(x-14)(x+1)$$

$x = 14$ $\quad$ $x = -1$ $\quad$ $\therefore x = 14, -1$

solve rational inequality

Solve for x using Factor Table $\frac{x+5}{x-3} \geq \frac{x-2}{x+1}$

Set equation = to zero

$$\frac{x+5}{x-3} - \frac{x-2}{x+1} \geq$$

$$\frac{x+5(x+1)}{(x-3)(x+1)} - \frac{(x-2(x-3))}{(x-3)(x+1)}$$

LCD : $(x-3)(x+1)$
$x \geq 0$
$x \neq 3, -1$
Important factors

① Simplify into standard form

$$\frac{x^2+x+5x+5-(x^2-3x-2x+6)}{(x-3)(x+1)} \geq 0$$

don't put into standard form

$$\frac{x^2+6x+5-(x^2-5x+6)}{(x-3)(x+1)} \geq 0$$

simplify

$$\frac{x^2+6x+5-x^2+5x-6}{(x-3)(x+1)} \geq 0$$

$$\frac{11x-1}{(x-3)(x+1)} \geq 0$$

→ x-int $(x = \frac{1}{11})$

$x \neq 3 \quad x \neq -1$ ~ x restrictions

② Put all x-values into factor table

	$-\infty$	-2	-1	0	$1/11$	2	3	4	∞
$x-3$	−			−		−		+	
$x+1$	−		+		+		+		
$11x-1$	−		−		+		+		
overall	−		+		−		+		

Positive between

$x \in (-1, 1/11] \cup (3, \infty)$

Round because restriction isn't included

is included

Word problems

A program generates values.

Taylor's Values: $T(x) = \frac{1}{x-2}$

Katy's value : $K(x) = \frac{2}{x+3}$

When will taylor's value be less than Katy's?

$$\frac{2}{x+3} > \frac{1}{x-2}$$

$$0 > \frac{1}{x-2} - \frac{2}{x+3}$$

LCD : $(x-2)(x+3)$

$$0 > \frac{x+3-2x+4}{(x-2)(x+3)}$$

$$0 > \frac{-x+7}{(x-2)(x+3)}$$

VA = −3, 2

x-int = $0 = -x+7$
$x = 7$
$(7, 0)$

	$-\infty$	-4	-3	1	2	4	7	8	∞
$-x+7$		+		+		+		−	
$x-2$		−		−		+		+	
$x+3$		−		+		+		+	
overall		+		−		+		−	

∴ Taylor's value will be less than Katy's value when $x \in [-3, 2) \cup (7, \infty)$

Make sure to use correct bracket

don't put R

Taegan can paint $3m^2$ wall in x hours
Natalie can paint $5m^2$ wall in $x+2$ hours
Together they can paint $11m^2$ in 3 hours
How long does it take each of them to pain their walls?

space ← $\frac{3}{x} + \frac{5}{x+2} = \frac{11}{3}$ > total

how long it takes. ←

Put unknown in denom

Focus on this portion and simplify.

$$= \frac{3(x+2) + 5(x)}{(x+2)(x)}$$

$$= \frac{3x+6+5x}{(x+2)(x)}$$

$$= \frac{8x+6}{(x+2)(x)} = \frac{11}{3}$$

Cross multiply denomenators

$$3(8x+6) = 11(x+2)(x)$$

$$24x+18 = 11x^2+22x$$

Simplify and set to zero

$$0 = 11x^2-2x-18$$

Use quadratic Formula to find x

$$\frac{1 \pm \sqrt{199}}{11} = \begin{array}{l} x = 1.37 \\ \cancel{x = -1.92} \\ x > 0 \end{array}$$

Teagan takes $x = 1.37$ hours

Natalie takes $x+2 = 1.37+2 = 3.37$ hours.

Chapter 4 – Rates of Change

ARC / EIRC , h = 0.01 AND word problems

ARC: $\dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$ **EIRC:** $\dfrac{f(a+h) - f(a)}{h}$

A ball is thrown and it's height H(t) is above level ground at time (t) units is modelled by $H(t) = 15t - 4t^2 + 18$. $H(t) = -4t^2 + 15t + 18$ rearranged

From what height is the ball thrown?

Make your $x = 0$

$H(0) = 18m$ ∴ Ball is thrown at 18metres.

When does the ball strike the ground?

set $y = 0$, use quadratic formula ($x \geq 0$)

$0 = -4t^2 + 15t + 18$

$\dfrac{-15 \pm \sqrt{513}}{-8} = \begin{cases} x = 4.71 \text{ sec} \\ x = -0.95 \text{ sec} \end{cases}$

∴ Ball strikes ground at approx 4.71 seconds.

What is the ARC of the ball from 0.72 sec (x_1) → 2.24 sec. (x_2)

Formula

$\dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$

① Plug in x-values in equation

$f(0.72) = 26.7264$

$f(2.24) = 31.5296$

② Plug into equation

$= \dfrac{31.5296 - 26.7264}{2.24 - 0.72}$

$= 3.16$

③ Therefore statement

∴ The ARC from 0.72 sec to 2.24 sec is $3.16 \dfrac{\text{metres}}{\text{sec}}$ (3.16 m/s)

When is the instentanious ROC at 3.47 seconds ($x = a$)

Formula

$\dfrac{f(a+h) - f(a)}{h}$

$h = 0.01$ always

① Plug in values for (a+h) & (a)

$f(a) = 21.8864$

$f(a+h) = 21.7584$

② Plug into equation

$\dfrac{21.7584 - 21.8864}{0.01}$

$= -12.8$

③ Therefore

∴ The EIRC at the height of ball at 3.47s is $-12.8 \dfrac{\text{metres}}{\text{sec}}$.

When does the ball reach a height of 15.56

$15.56 = -4t^2 + 15t + 18$

set equation to $y = 0$ and solve

$0 = -4t^2 + 15t + 2.44$

use quad formula

$\dfrac{-15 \pm \sqrt{264.04}}{-8} = \begin{cases} x = -0.156 \text{ – no negative time} \\ x = 3.906 \end{cases}$

∴ The ball reaches this height at ≐ 3.91 sec

How to find max height

$\max = \dfrac{-b}{2a}$ (sec)

A sphere shaped balloon is being filled at a constant rate. The volume of the sphere is $V(t) = \frac{4\pi t^3}{3}$ t - time in minutes, V - volume in cm^3

① What is the volume after 6 minutes

Plug in 6 for (t)

$V(6) = \frac{4\pi(6)^3}{3}$ $\quad V(6) = 288\pi$ $(= 964.78)$

∴ The volume of the balloon after 6 min is $288\pi\ cm^3/min$ or APPROX $964.78 cm^3/min$.

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② What is ARC between $2 \le t \le 4$

$V(2) = \frac{32}{3}\pi\ (33.5)$

$V(4) = \frac{256}{3}\pi\ (268.1)$

$\frac{V(t)_2 - V(t)_1}{t_2 - t_1}$

$\frac{\left(\frac{256}{3}\pi\right) - \left(\frac{32}{3}\pi\right)}{4-2}$ → $\doteq 117.2861257$

∴ The ARC of the volume from $2 \le t \le 4$ is approx $117.29 cm^3/min$.

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③ What is elapsed time when volume is $75 cm^3$

$75^{\times 3} = \frac{4\pi t^3}{3}^{\times 3}$ — isolate (t)

$\frac{\sqrt[3]{225}}{4\pi} = \frac{\sqrt[3]{4\pi t^3}}{4\pi}$

$t = \frac{\pm\sqrt[3]{225}}{4\pi}$ — always make sure to add ± before root

$x = 2.616$

$x = -4.84$ → no negative time

∴ Therefore elapsed time when $V = 75 cm^3$ is 2.62 min.

Water drains out of a 2000L tub, amount of water remaining is rep by $A(t) = 0.00002(100-t)^4$ t = elapsed time in min, $0 \le t \le 100$

$A(t)$ = is water remaining in litres

① What is ARC in the last 15 min?

$100 - 15 = 85$ min

- 85 min passed so ARC is between $100 \le t \le 85$

$f(100) = 0$

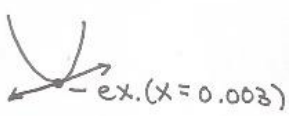
$f(85) = 1.0125$

$\frac{A(t)_2 - A(t)_1}{t_2 - t_1}$

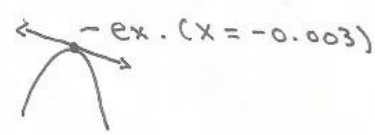
$\frac{1.0125 - 0}{85 - 100}$ → ARC = -0.0675

∴ Therefore ARC in the last 15 min is -0.07 L/min.

Using EIRC, you can figure out if your answer will be a max or min.



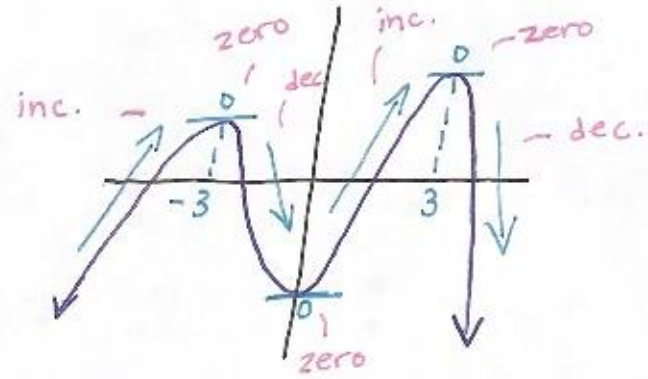
If EIRC answer is a small positive number, the point is a minimum



If EIRC is a small negative number, the point is a maximum.

intervals where rates of change are - / 0 / +

State when EIRC for f(x) is zero, positive, negative



zero

$\boxed{x = -3, 0, 3}$

positive

$\boxed{(-\infty, 0) \cup (0, 3)}$

Negative

$\boxed{(-3, 0) \cup (3, \infty)}$

equation of a tangent line

Find equation of tangent line (using EIRC)

Calculate EIRC at the function $x = 4$

$f(x) = \frac{6x}{3x-5}$

Restric. $= x \neq \frac{5}{3}$

$f(\overset{4}{x}) = \frac{24}{7}$

$f(\underset{4+0.01}{x+h}) = \frac{2406}{703}$

① Find EIRC

$\frac{f(x+h) - f(x)}{h}$

$\frac{\left(\frac{2406}{703}\right) - \left(\frac{24}{7}\right)}{0.01} \longrightarrow = \frac{-3000}{4921} \doteq -0.6096$

② Find tangent line

-use the equation $y = mx + b$ to solve

Given:

$x = 4$

$y = \frac{24}{7}$ y-value of (-4)

$m = -0.6096$ EIRC

$y = mx + b$

$\frac{24}{7} = -0.6096(4) + b$ move and simplify

$\frac{24}{7} + 2.4385 = b$

$b \doteq 5.87$

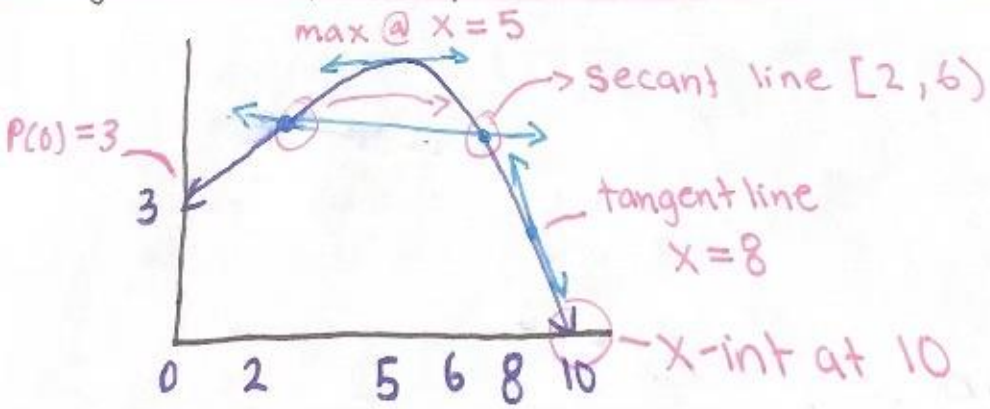
Now re-write $y = mx + b$ in correct format

only m and b value ← $y = -0.61x + 5.87$

Other

Draw simple parabolic curve where $P(0) = 3$, x-int at 10, max at $x = 5$

Draw a tangent line at $x = 8$, secant line on interval $[2, 6]$. x_1, x_2

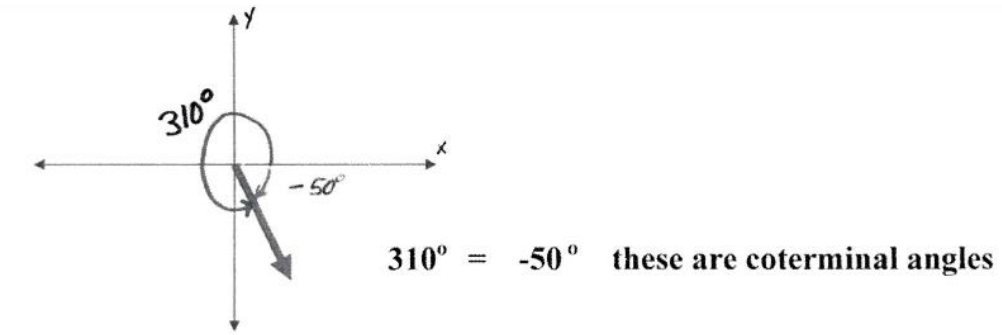


Chapter 5 – Trigonometric Functions

radians/degrees conversion

$$\frac{degrees}{radians} = \frac{180}{\pi} \qquad \frac{radians}{degrees} = \frac{\pi}{180}$$

related / co-related angles



310° = -50° these are coterminal angles

transformations of sine/cosine curves

$$y = \mathrm{a}\,\boldsymbol{sin}\big(\mathrm{b}(x-\theta)\big) + \mathrm{k}$$

a = | a | (amplitude)

- If |**a**| is negative = reflection in the x axis
- If |**a**| is a decimal/less than 1 = vertical compression by a factor of $\frac{1}{|a|}$ OR the denominator of the given fraction
- If |**a**| is greater than 1 = vertical stretch by a factor of |**a**|

b = | b | $\left(\frac{360°}{period} = b\right)$

- If |**b**| is negative = reflection in the y axis
- If |**b**| is a decimal or fraction = horizontal stretch by a factor of $\frac{1}{|a|}$ OR the denominator of the given fraction
- If |**b**| is greater than 1 = horizontal compression by a factor of |**a**|

θ = phase shift (horizontal)

- If **θ** is greater than 1 = phase shift **θ** radians/degrees right
- If **θ** is less than 1 = phase shift **θ** radians/degrees left

k = phase shift (vertical)

- If **k** is greater than 1 = vertical translation **k** units up
- If **k** is less than 1 = vertical translation **k** units down

$g(x) = \frac{1}{5}\cos[4x + \frac{3\pi}{2}] - \sqrt{2}$ $= \frac{1}{5}\cos[4(x + \frac{3\pi}{8})] - \sqrt{2}$	• vertical compression of factor 5 • horizontal compression of factor 4 • phase shift $\frac{3\pi}{8}$ radians left • vertical translation $-\sqrt{2}$ units down

$f(x) = -4\sin[2(x - \frac{\pi}{9})] + 3$	• reflection in the x axis • vertical stretch of factor 4 • horizontal compression of factor 2 • phase shift $\frac{\pi}{9}$ radians right • vertical translation 3 units up

$y = 7\sin[-\frac{1}{2}x - \frac{\pi}{7}] + 13$ $= 7\sin\left[-\frac{1}{2}\left(x + \frac{2\pi}{7}\right)\right] + 13$	• vertical stretch of factor 7 • reflection in the y-axis • horizontal stretch of factor 2 • phase shift $\frac{2\pi}{7}$ radians left • vertical translation 13 units up

graph a transformed sine/cosine function

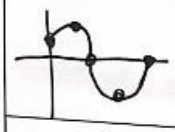
...with a Phase Shift

Factor out K

EX. 6. Graph one cycle in radians of $f(x) = 3\sin\left(4x - \frac{5\pi}{3}\right) + 8$.

$4\left(x - \frac{5\pi}{12}\right)$

sin

Parent	Child
(x, y) 	$\left(\frac{x}{k} + d,\ ay + c\right)$ $\left(\frac{x}{4} + \frac{5\pi}{12},\ 3y + 8\right)$
$(0, 0)$	$\left(\frac{5\pi}{12},\ 8\right)$
$\left(\frac{\pi}{2}, 1\right)$	$\left(\frac{13\pi}{24},\ 11\right)$
$(\pi, 0)$	$\left(\frac{2\pi}{3},\ 8\right)$
$\left(\frac{3\pi}{2}, -1\right)$	$\left(\frac{19\pi}{24},\ 5\right)$
$(2\pi, 0)$	$\left(\frac{11\pi}{12},\ 8\right)$

H. Axis = $f(x) = C$

$f(x) = 8$

new c - line

period $= \frac{2\pi}{4}$

Period $= \frac{\pi}{2}$

amplitude $= |a|$

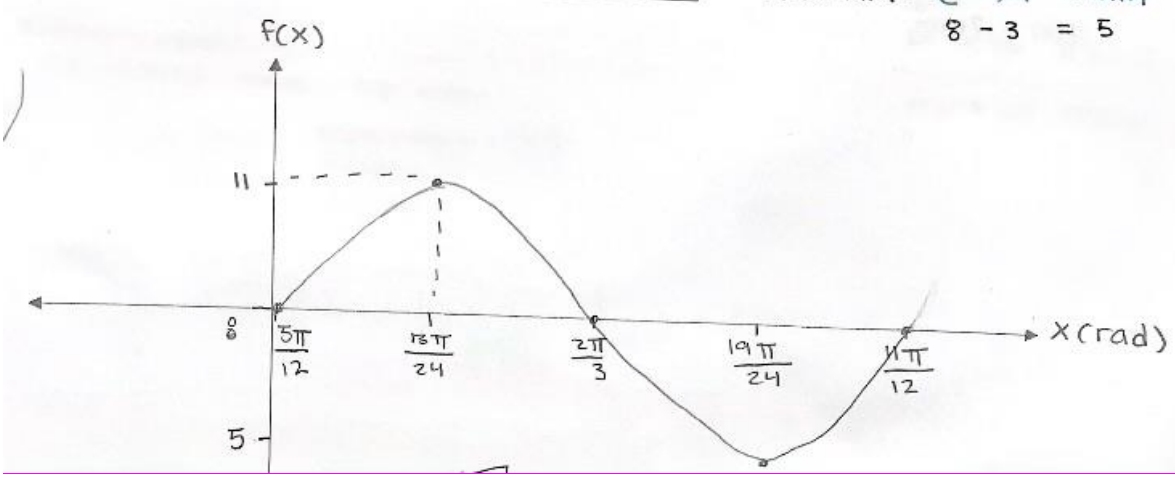
$= 3$

Find Max = $C + A = \text{Max}$

$8 + 3 = 11$

Find Min = $C - A = \text{min}$

$8 - 3 = 5$



EX. 7. Graph one cycle in radians of $f(x) = -4\cos\left(\frac{1}{3}x + \frac{2\pi}{7}\right) + 1$. $\frac{1}{3}\left(x + \frac{6\pi}{7}\right)$

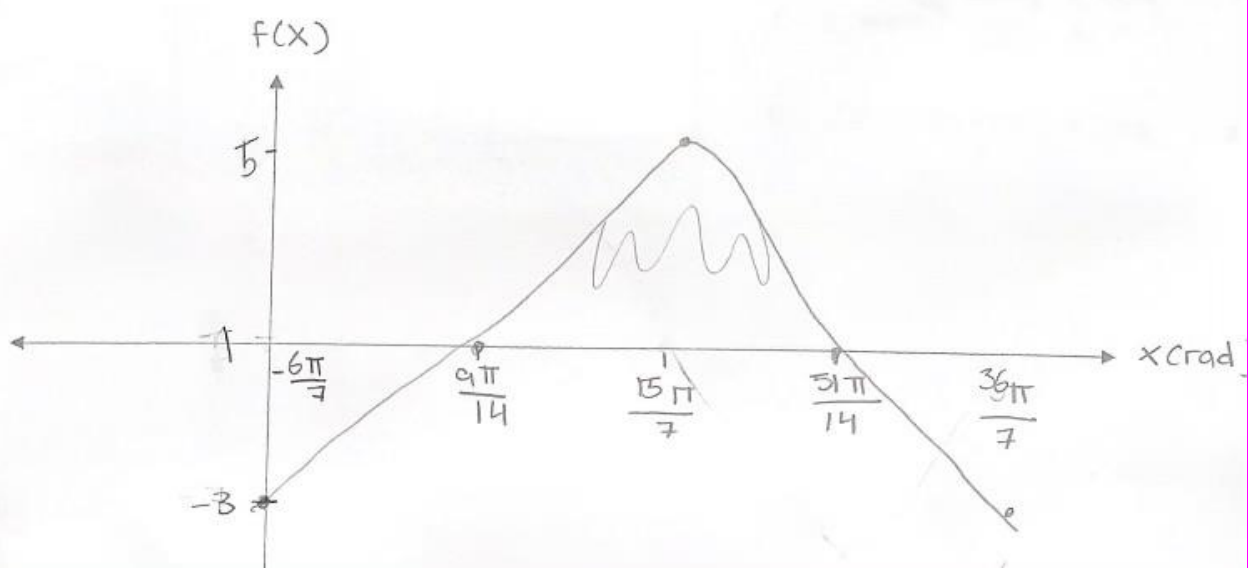
Parent (x, y)	Child $\left(\frac{x}{K} + d,\ ay + c\right)$ $\left(\frac{x}{1/3} - \frac{6\pi}{7},\ -4y + 1\right)$
$(0, 1)$	$\left(-\frac{6\pi}{7}, -3\right)$
$\left(\frac{\pi}{2}, 0\right)$	$\left(\frac{9\pi}{14}, 1\right)$
$(\pi, -1)$	$\left(\frac{15\pi}{7}, 5\right)$
$\left(\frac{3\pi}{2}, 0\right)$	$\left(\frac{51\pi}{14}, 1\right)$
$(2\pi, 1)$	$\left(\frac{36\pi}{7}, -3\right)$

H Axis $= 1$

Period $= \frac{2\pi}{K}$

$= 6\pi$

Amp $= 4$



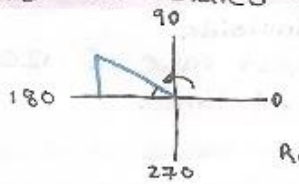
equivalent expressions using the related angle

Related and Co-related angles

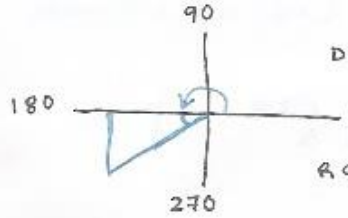
θ - principal angle

R - related angle

C - co-related angle



Degrees	$R = 180 - \theta$ $C = 90 - R$
Radians	$R = \pi - \theta$ $C = \pi/2 - R$



Degree	$R = \theta - 180$ $C = 90 - R$
Radians	$R = \theta - \pi$ $C = \pi/2 - R$

$\theta = 180 + R$

$\theta =$	$R = 360 - \theta$ $C = 90 - R$
$R =$	$R = 2\pi - \theta$ $C = \pi/2 - R$

Equivalent related and co-related

$\sin R = \cos C$
$\sin R = \cos(90 - R)$

$\sec R = \csc C$
$\sec R = \csc(90 - R)$

$\tan R = \cot C$
$\tan R = \cot(90 - R)$

Ex. $\sin \frac{\pi}{8}$ as an equivalent cos expression

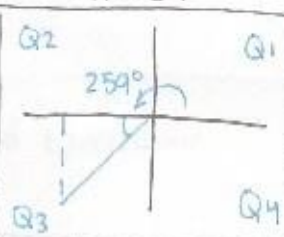
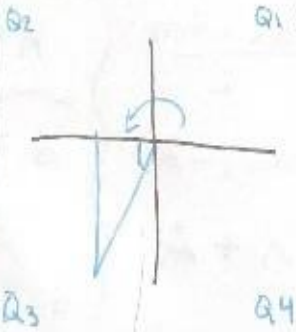
$$\sin \frac{\pi}{8} = \cos\left(\frac{\pi}{2} - \frac{\pi}{8}\right) = \cos \frac{3\pi}{8}$$

① Find principal angle = Related angle

② If principal angle is larger than 180, do $(180 - \theta)$

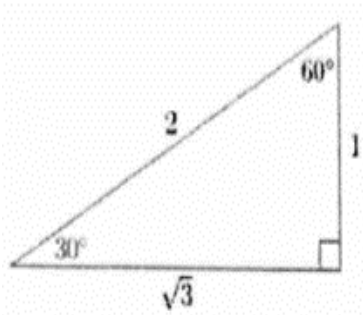
③ Find principal angle = co-related angle

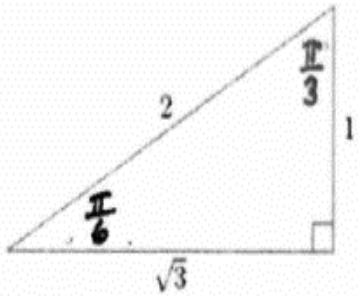
④ Check answers w/ calculator

Ex.	Sketch	Related ∠	Co-Related ∠
$\tan 259°$ (+) in Q3		$\tan\theta = \tan R$ $\tan 259° = \tan 79°$ $(259 - 180)$	$\tan\theta = \cot C$ $\tan 259 = \cot(90 - 79)$ $\tan 259° = \cot 11°$
$\sec \frac{15\pi}{11}$ (245.45) In Q3 (−sec)		$\sec\theta = -\sec R$ $\sec \frac{15\pi}{11} = -\sec \frac{4\pi}{11}$ $\left(\frac{15\pi}{11} - \pi\right)$	$\sec\theta = -\csc C$ $\sec \frac{15\pi}{11} = -\csc \frac{3\pi}{22}$ $\left(\frac{\pi}{2} - \frac{4\pi}{11}\right)$

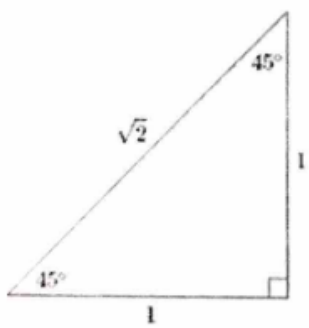
special triangles, exact values

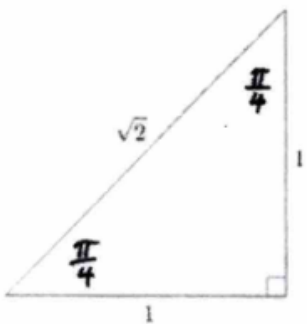
- **The 30-60-90 Triangle**
  Degree Form Radian Form





- **The 45-45-90 Triangle**
  Degree Form Radian Form





Arc length

A circle has a central angle $\frac{3\pi}{2}$ rad. If arc length is 2.375 cm, what is radius as an exact value?

G: 2.375 cm $= \frac{19}{8}$ cm

$\theta R = \frac{3\pi}{2}$ rad

$r = \frac{a}{\theta R}$

$r = \frac{19}{8} \div \frac{3\pi}{2}$

$\frac{19}{8} \times \frac{2}{3\pi}$ - Flip and multiply

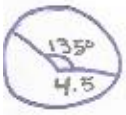
$= \frac{38 \div 2}{24\pi \div 2} = \frac{19}{12\pi}$ reduce

∴ Therefore radius is $\frac{19}{12\pi}$ cm

Unit 5 A circle of radius 4.5 cm has a central angle of 135°. Calculate arc length.

Given: $\theta R = 135°$

$r = 4.5$ cm



① Change to radians

$135° = \frac{135\pi}{180}$

$= \frac{3\pi}{4}$ rad

② Find arc length

$a = r \times \theta R$

$a = (4.5)\left(\frac{3\pi}{4}\right)$

$a = \frac{27\pi}{8}$

∴ Arc length is $\frac{27\pi}{8}$ rad.

Word Problems

A ferris wheel completes 3 revolutions every 38 min, the max height is 95 m. The ride starts from the lowest point of 2m above ground.

$f(x) = a\sin[K(x+d)] + c$

- express as a sine function

- 3 rev every 38 min $= 38 \div 3 = \frac{38}{3} \rightarrow K = 2\pi \div \frac{38}{3} = \frac{6\pi}{38}$ REDUCE $\rightarrow \frac{3\pi}{19}$

- max height is 95m, min height is 2m. Use that to find A and C

$$A = \frac{95-2}{2} = \frac{93}{2} \qquad C = \frac{95+2}{2} = \frac{97}{2}$$

- to find (d), multiply period by $\left(\frac{1}{4}\right)$ or divide by 4, since it reaches max $\frac{1}{4}$ of its cycle
OR divide (K)-value by $90 / \frac{\pi}{2}$

Period $= \frac{38}{3}$

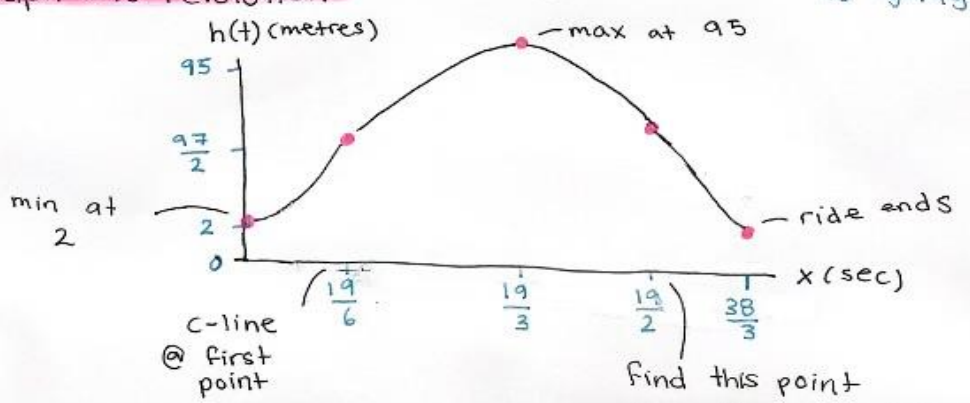
K value $= \frac{6\pi}{38}$

① $\frac{38}{3} \times \frac{1}{4} = \frac{19}{6}$ ② $\frac{\pi}{2} \div \frac{3\pi}{19} = \frac{19}{6}$

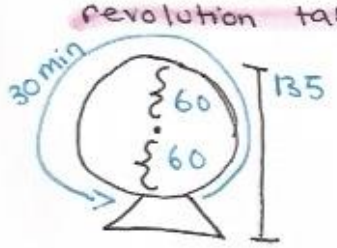
- Plug in values into equation $\rightarrow f(x) = \frac{93}{2}\sin\left[\frac{3\pi}{19}\left(x - \frac{19}{6}\right)\right] + \frac{97}{2}$

negative because graph is going to the right.

Graph one revolution



A ferris wheel has a diameter of 120m and a Max height of 135m. One revolution takes 30 min.



Max height = 135

Radius (a value) $= 120 \div 2 = 60$

C line $= 135 - 60 = 75$

Min value $= C - A = 15$

d value $=$ period $\div 4 = 30 \div 4 = \frac{15}{2}$

K value $= 2\pi \div$ Period $= 2\pi \div 30 = \frac{\pi}{15}$

$$h(x) = 60\sin\left[\frac{\pi}{15}\left(x - \frac{15}{2}\right)\right] + 75$$

How many rev are completed in 13 min

$$\frac{x}{13 \text{ min}} = \frac{1 \text{ rev}}{30 \text{ min}} \qquad x = \frac{13}{30} \text{ rev}$$

Chapter 6 – Trigonometric Equations and Identities

solve trig equation (linear or quadratic)

LINEAR TRIG EQUATIONS

- When finding both values of sin, subtract related angle by pi (180)
- When finding values of cosine, subtract related angle by 2pi (360)
- When finding values of tan, add pi (180)
- When having a negative value, add 2pi (360) to find it on the Cartesian plane. Treat its related angle as a positive.
- When A value is not 1, divide it by the denominator of the length given to you to remove it
- When you have inverse trig ratios, change it to its opposite
- When you have a phase shift, solve without the shift and add it later

EX 11. Solve $2\sin 3x = \sqrt{3}$ $\quad 0 \le x \le 360$ *degrees*

$$2\sin 3x = \sqrt{3}$$

$$\sin 3x = \frac{\sqrt{3}}{2}$$

$$3x = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$x = \frac{\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)}{3}$$

$$p = \frac{360^\circ}{b}$$

$$p = \frac{360^\circ}{3}$$

$$p = 120^\circ$$

π/k - d

$$x = \frac{180^\circ}{3} - 20^\circ$$

✓ $x = 40^\circ$

$+ 120^\circ$

✓ $x = 160^\circ$

$+ 120^\circ$

✓ $x = 280^\circ$

or

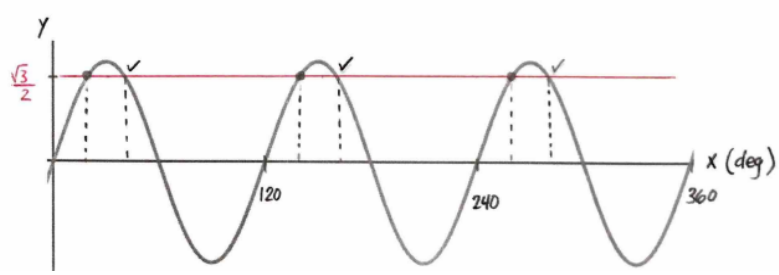
- $x = 20^\circ$

$+ 120^\circ$

- $x = 140^\circ$

$+ 120^\circ$

- $x = 260^\circ$



- Move 2 into denom and put inverse of sin
- Divide by 'k' on both sides
- Find the period, it will help find the angles
- Add period to the angle you get to get more answers
- To get other side continue doing 180-R but divide all by 'k' value which is 3

$5\cot x + \sqrt{8} = 0$ on $[0, 2\pi]$ – answers have to be between $0 \to 6.28$

$5\cot x + \sqrt{8} = 0$

$5\cot x = -\sqrt{8}$ – move $\sqrt{8}$

$\frac{\cot x}{5} = \frac{-\sqrt{8}}{5}$ – divide both sides by 5

$\frac{1}{\tan x} = \frac{-\sqrt{8}}{5}$ – change to inverse of $\cot x$

$\tan x = -\frac{5}{\sqrt{8}}$ – flip the fraction on both side

$(\tan^{-1})\, x = \tan^{-1}\left(-\frac{5}{\sqrt{8}}\right)$ – multiply both sides by inverse

$x \doteq -1.06$ – can't be an answer $x > 0$

Find values of x

$x = -1.06$

$+\pi$

$x = 2.08$

1 cycle of tan is also $(+\pi)$

– cycle of sin is $(+2\pi)$

– cycle of cos is $(+2\pi)$

OR

$x = \pi + (-1.06)$

$\doteq 2.08$

$x = 2.08 + \pi$

$\doteq 5.22$

Period of tan is $(+\pi)$ so add π to each

– Period of sin is $(\pi - x)$

– Period of cos is $(2\pi - x)$

$2\sin 3x = \sqrt{3}$

- If there is a k value greater than 1
- Cycle of sin/cos is not $+2\pi$
- You must divide 2π by k to get new cycle value
- You must also do $\frac{\pi}{k} - d$ for period

$\frac{2\pi}{3}$ $(120°)$

$\frac{\pi}{3} - d$ – d-value is the answer you get from simplifying solution

$\frac{x = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)}{3} = 20°\left(\frac{\pi}{9}\right)$

Trig equations

$\sin^2 x + \sin x = 0$ factor out a $\sin x$ Between $0 \le x \le 2\pi$

$\sin x(\sin x + 1) = 0$

$\sin x = 0$ | $\sin x + 1 = 0$

$x = \sin^{-1}(0)$ | $x = \sin^{-1}(-1)$

$x = 0$ | $x = \pi - 0$ | $x \doteq -1.57$

$+2\pi$ $x = 6.28$ OR $x = 3.14$ $x = 4.71$ 2π

$x \doteq \{0, 3.14, 4.71, 6.28\}$ rad

$\frac{3\sin^2 x}{3\cos^2 x} = \frac{2\cos^2 x}{3\cos^2 x}$ – divide both sides by $3\cos^2 x$

$\frac{\sin^2 x}{\cos^2 x} = \frac{2}{3}$ – change $\frac{\sin}{\cos} = \tan$

$\sqrt{\tan^2 x} = \sqrt{\frac{2}{3}}$ – square root to remove x^2

$\tan x = \pm\sqrt{\frac{2}{3}}$

$x = \tan^{-1}\left(\sqrt{\frac{2}{3}}\right)$ | $x = \tan^{-1}\left(-\sqrt{\frac{2}{3}}\right)$

$x \doteq 0.68$ $+\pi$ (coterminal angle) | $x \doteq -0.68$ | OR $x = \pi + (-0.68)$

$x \doteq 3.82$ | $+\pi = 2.46$ | $\doteq 2.46$

OR tan, you ($+\pi$) period | $+\pi = 5.60$

$x \doteq \pi + 0.68$

$= 3.82$

$\therefore x \doteq \{0.68, 2.46, 3.82, 5.60\}$ rad – don't forget this

$10\sin^2 x - \sin x = 2$ $0 \le x \le 5$ rad

$10\sin^2 x - \sin x - 2 = 0$ – set equation to zero

(Let A rep $\sin x$) –> $10A^2 - 1A - 2 = 0$ – let A rep $\sin x$ to make it easier

$10A^2 - 5A + 4A - 2 = 0$ – factor

$5A(2A-1) + 2(2A-1) = 0$

$(5A+2)(2A-1)$

Plug in $\sin x$ again

$(5\sin x + 2)$ $(2\sin x - 1)$

find x

$\sin x = -\frac{2}{5}$ | $x = \sin^{-1}\left(\frac{1}{2}\right)$

$x = \sin^{-1}\left(\frac{-2}{5}\right)$ | $x = \frac{\pi}{6}$ (0.52)

$x = -0.41$ $+2\pi$ | $x = \pi - 0.52$

$x \doteq 5.87$ – needs to be less than 5 | $= 2.62$ $+2\pi$

subtract by period | $= 8.90$

$x = \pi - (-0.41)$

$= 3.55$ $+2\pi$

add by cycle $= 9.83$

$x \doteq \{0.52, 2.62, 3.55\}$ rad

trig identity

related / corelated angles

Determine exact value of $\cos \frac{11\pi}{12}$ $(165°)$

- cos is negative in the 2nd quad

- Find co-terminal angle

$$\pi - \frac{11\pi}{12} = \frac{\pi}{12} (15°)$$

- Find 2 angle that add/sub to that

$$\frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \cos\frac{\pi}{4}\cos\frac{\pi}{6} + \sin\frac{\pi}{4}\sin\frac{\pi}{6}$$

$$\cos\frac{\pi}{12} = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$$

$$\cos\frac{\pi}{12} = \frac{\sqrt{6}+\sqrt{2}}{4} \longrightarrow -\cos\frac{\pi}{12} = \boxed{\cos\frac{11\pi}{12} = \frac{-(\sqrt{6}+\sqrt{2})}{4}}$$

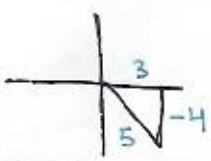
change to (−)

- If it was $\sec \frac{11\pi}{12}$ you would do the inverse of sec

$$\sec\frac{11\pi}{12} = -\frac{\sqrt{6}+\sqrt{2}}{4} = \frac{-1}{\frac{\sqrt{6}+\sqrt{2}}{4}} \longrightarrow \left(\sec\frac{11\pi}{2} = \frac{-4}{\sqrt{6}+\sqrt{2}}\right)$$

equivalent expressions

- If $\left(\tan -\frac{4}{3}\right)$ Find $\sin 2\theta$, $\cos 2\theta$, $\tan 2\theta$



$$hyp = \sqrt{4^2 + 3^2} = 5$$

Sin2θ

$\sin\theta = \frac{-4}{5}$ $\cos = \frac{3}{5}$

$$\boxed{\sin 2\theta = 2\sin\cos}$$

$$\sin 2\theta = 2\left(-\frac{4}{5}\right)\left(\frac{3}{5}\right)$$

$$\sin 2\theta = \frac{-24}{25}$$

Cos2θ

$$\boxed{\cos 2\theta = \cos^2\theta - \sin^2\theta}$$

$$= \left(\frac{3}{5}\right)^2 - \left(-\frac{4}{5}\right)^2$$

$$\cos 2\theta = \frac{-7}{25}$$

Tan2θ

$$\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta}$$

$$= \left(-\frac{24}{25}\right) \times \left(-\frac{25}{7}\right)$$

$$\tan 2\theta = \frac{24}{7}$$

$\therefore$ $\sin 2\theta = -\frac{24}{25}$ $\tan 2\theta = \frac{24}{7}$

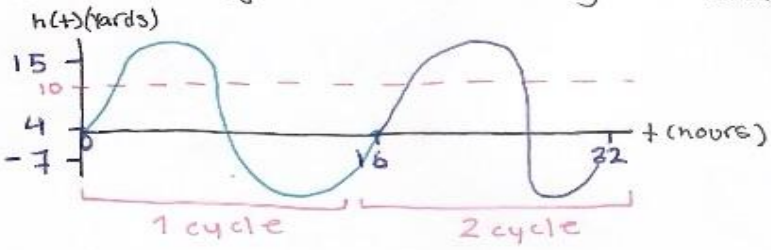
$\cos 2\theta = -\frac{7}{25}$

Word problems

The height is modelled by $h(t) = 11\sin[\frac{\pi}{8}(t-2)] + 4$

$h(t)$ – height in yards
t – time in hours

When is the object above 10m during the first two cycles?



- label your graph
- use $C \pm A$ to find max and min
- C value is middle line
- draw dotted line @ 10m
- find period by doing $\frac{2\pi}{\pi/8}$

$k = 16$ (1 cycle)

- First solve without the phase shift

$$\frac{10}{11} = 11\sin\frac{\pi}{8}t + 4$$

$$\frac{10-4}{11} = \sin\frac{\pi}{8}t$$

$$(\sin^{-1})\ \frac{6}{11} = \frac{\pi}{8}t\ (\sin^{-1})$$

$$\left(\frac{8}{\pi}\right)\sin^{-1}\frac{6}{11} = \frac{\pi}{8}t\left(\frac{8}{\pi}\right)$$

$$\left(\frac{8}{\pi}\right)\sin^{-1}\left(\frac{6}{11}\right) = t$$

$t = 1.47$ hrs OR $t = 8 - 1.47$
$= 6.53$ hrs

half a period because sin is $(\pi -)$

- Now add phase shift $(+2)$

$x = 1.47 + 2$ OR $x = 6.53 + 2$
$\doteq 3.47$ $\doteq 8.53$

Now add a cycle

$+16$ $+16$

$\doteq 19.47$ $\doteq 24.53$

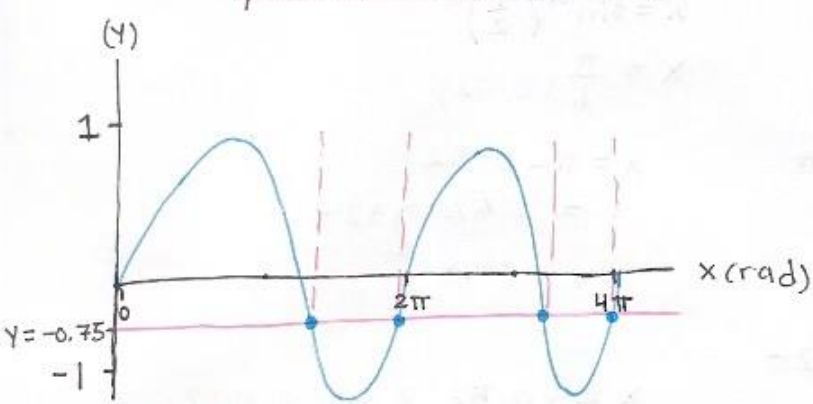
The object is above 10m on approx

$(3.47, 8.53)$ ~~hrs~~ $\cup$ $(19.47, 24.53)$ hrs

make sure to add time

★ same thing done with cos

$y = \sin x$ and $y = -0.75$



which Quad are solutions located in?

- sin in negative in the third and fourth quad

∴ Third and Fourth quadrant

$\sin^{-1}(-0.75)$

$x = -0.85$ $+2\pi$

① $x = 5.44$ $+2\pi$

② $x = 11.71$

OR

$x = \pi - (-0.85)$

③ $= 3.99$ $+2\pi$

④ $x = 10.27$

∴ 4 solutions

Chapter 7 - Exponential and Logarithmic Functions

graph log function

Y = a log $_B$ (b (x - h)) + c -> to find new points: $(\frac{x}{b} + h \quad , \quad ay + c)$

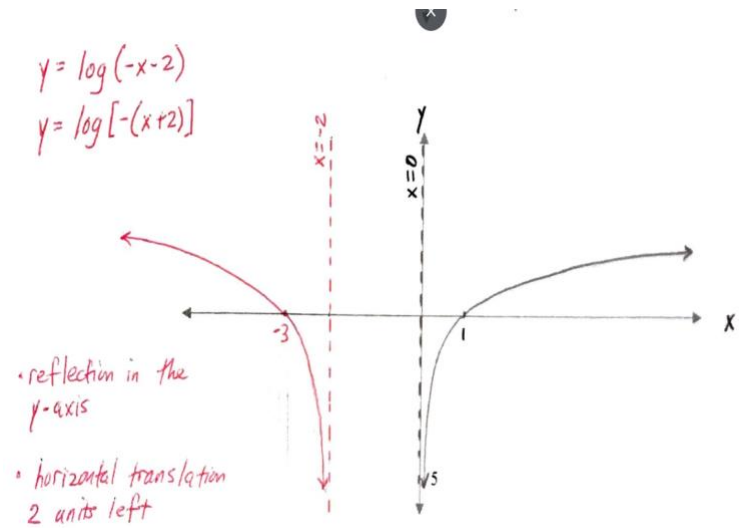
a = reflection in the x- axis or vertical stretch/compression

B = base of log

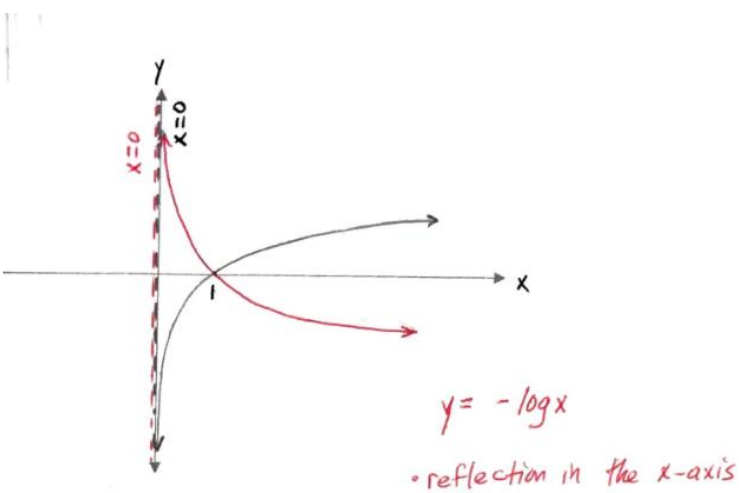
b = reflection in the y-axis or horizontal stretch/compression

h = horizontal translation left/right

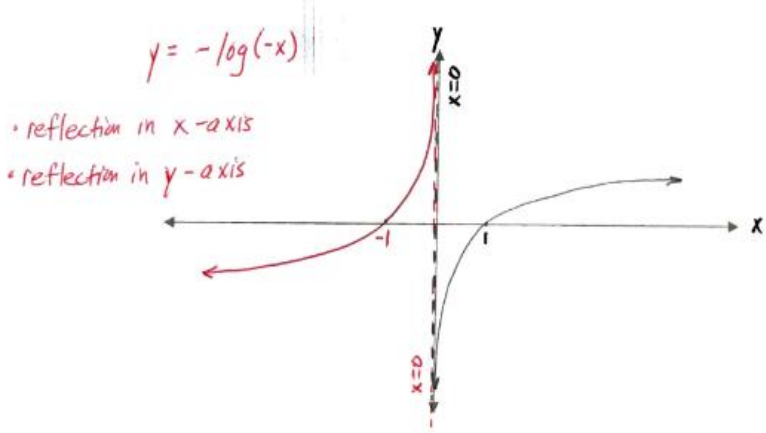
c = vertical translation up/down



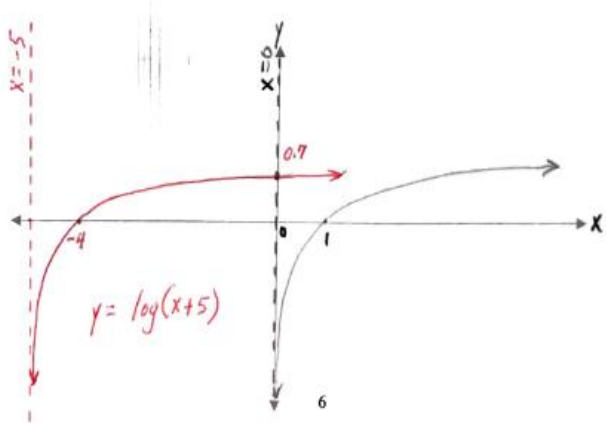
- Isolate x by removing (- 1)
- Vertical asymptote always at 0 for original, for transformed the 'd' value determines it, which is (-2) in this case
- Point will be one over from the asymptotes
- Since there is a reflection in the y-axis, the point for transformed function will be one behind asymptote instead of one ahead
- Name transformations
- Always remember to put x and y on graph



- Vertical asymptote at only 0 since there are no phase shifts
- Reflection in the x-axis which means it will be flipped
- BLACK IS ORIGINAL, RED IS TRANSFORMED
- Name transformations
- Put x and y on graph



- Vertical asymptote at only 0 since there are no phase shifts
- Reflection in the x-axis which means it will be flipped
- Reflection in the y- axis as well, must be flipped along the y line
- BLACK IS ORIGINAL, RED IS TRANSFORMED
- Name transformations
- Put x and y on graph



- BLACK IS ORIGINAL, RED IS TRANSFORMED
- Name transformations
- Put x and y on graph
- Vertical asymptote at 0 for original and -5 for transformed according to log function
- Point is one number ahead of asymptote, unless a reflection in y-axis

EX 4. Graph the parent function $y = \log x$ Graph $y = 3\log(-2x-8) + 1$. Use 1 cm : 2 units for both.

Parent	Child
(x, y)	$\left(\frac{x}{b} + h,\ ay + k\right)$
$y = \log x$	$\left(\frac{x}{-2} - 4,\ 3y + 1\right)$
$(\frac{1}{10}, -1)$	$\left(-\frac{81}{20},\ -2\right)$
$(1, 0)$	$\left(\frac{-9}{2},\ 1\right)$
$(10, 1)$	$(-9,\ 4)$

$y = 3\log[-2(x+4)] +$

$a = 3$ $b = -2$

$h = -4$ $k = 1$

- vertical stretch of factor 3
- reflection in y-axis
- horizontal translation 4 units left
- vertical translation 1 unit up.

- Don't need to graph
- Factor out B value
- Used mapping formula to find new points
- MEMORIZE PARENT LOG POINTS

simplify log expressions

$\log_3 \frac{1}{81} = y$

$3^y = \frac{1}{81}$

$3^y = \frac{1}{3^4}$

$3^y = 3^{-4}$

$y = -4$

- 3 is the base
- 1/81 is the "argument"
- Make the base equal on both sides
- Now do a zoom up on the exponents and focus on exponents only
- Evaluate.
- Y = -4

$\log_5 \sqrt[3]{25} = y$

$5^y = \sqrt[3]{25}$

$5^y = (25)^{\frac{1}{3}}$

$5^y = (5^2)^{\frac{1}{3}}$

$5^y = 5^{\frac{2}{3}}$

$y = \frac{2}{3}$

- 5 is the base
- 3 root 25 is the argument
- Change the argument into exponent form (writing the room is just 1/x)
- Make base equal on both sides
- Multiply exponants in the brackets to outside
- Y = 2/3

$\log_2(-4) = y$

$2^y = -4$

$2^y > 0$ for all real y

there is no solution

$\log_5 \frac{1}{25} = -2$

$5^{-2} = \frac{1}{25}$

- Find exponential form
- Base is 5
- Exponent is Y value, which is -2
- And argument is 1/25

$\log_5 1 = y$

$5^y = 1$

$5^y = 5^0$

$y = 0$

- 5 is the base
- When argument is one, you can change the base to its equal and put "0" as the exponent. Anything to the power of '0' is 1
- Except zero to the power of zero

$\log_{\frac{1}{4}} x = -2$

$\left(\frac{1}{4}\right)^{-2} = x$

$(4^{-1})^{-2} = x$

$4^2 = x$

$16 = x$

- 1/4 is the base
- -2 is the exponent
- X is argument
- Put it into exponent form
- Flip 1/4 to turn remove fraction
- Evaluate for X
- X = 16

a. $\log_x 64 = 3.$

$x^3 = 64$

$(x^{\cancel{3}})^{\frac{1}{\cancel{3}}} = 64^{\frac{1}{3}}$

$x = 4$

- Base is X
- Exponent is 3
- Argument is 64
- Get rid of exponent on x by doing 1/3 on both sides
- X= 4

$\log_{25} \sqrt[3]{5} = y$

- 25 is base
- Y is exponent
- 3 root 5 is argument
- Change both to same base

$25^y = \sqrt[3]{5}$

$(5^2)^y = 5^{\frac{1}{3}}$

- Make 25 into 5^2 and change 3 root 5 into $5^{1/3}$

$5^{2y} = 5^{\frac{1}{3}}$

- Multiply exponents together to make 2y

$2y = \frac{1}{3}$

- Solve exponents

$y = \frac{1}{6}$

$6 \log_3 27) = y$

- Solve in log form
- Divide both sides by 6

$\log_3 27 = \frac{y}{6}$

- Change to exponent form

$3^{\frac{y}{6}} = 27$

- Make bases the same

$3^{\frac{y}{6}} = 3^3$

- Solve exponents

$\frac{y}{6} = 3$

$y = 18$

solve exponential / log equations

1. $9^{2x+1} = 81(27^x)$

$(3^2)^{2x+1} = 3^4(3^{3x})$ - make same base

$3^{4x+2} = 3^{3x+4}$ since base is same add the exponants. 4 cannot be merged to 3x which is why there is a (+).

$4x+2 = 3x+4$

$x = 2$ calculate (x)

2. $2^{x+2} + 2^x = 320$

$(2^x \times 2^2) + 2^x = 320$ Imagine it written like this

$2^x(2^{x+2} + 2x) = 320$

$2^x(4+1) = 320$ factor out 2^x

$\frac{2^x(5)}{5} = \frac{320}{5}$ simplify exponents

$2^x = 64$ divide to isolate (x)

$2^x = 2^6$ - find (x)

$x = 6$

3. $\left(\frac{1}{4}\right)^{2x+1} = \sqrt{8}$ remove root and fraction

$(4^{-1})^{2x+1} = 8^{\frac{1}{2}}$

$((2^2)^{-1})^{2x+1} = (2^3)^{\frac{1}{2}}$ - same base

$2^{-4x-2} = 2^{\frac{3}{2}}$ - evaluate exponants

$-4x-2 = \frac{3}{2}$ - move (-2)

$\frac{-4x}{-4} = \frac{\frac{7}{2}}{-4}$ - divide to isolate (x)

$x = -\frac{7}{8}$

4. $2^{x+1} = 3^{x-1}$ change to log and put exponat at front

$\log 2^{x+1} = \log 3^{x-1}$

$\frac{(x+1)\log 2}{\log 2} = \frac{(x-1)\log 3}{\log 2}$ - divide by log 2

$(x+1) = (x-1)\,1.58496$ — multiply into answer

$(x+1) = 1.58496x - 1.58496$ move to keep (x) positive

$1 + (1.58496) = 1.5846x - 1x$

$\frac{2.58496}{0.5846} = \frac{0.5846x}{0.5846}$ divide to isolate (x)

$x \doteq 4.42$

Word Problems

Growth and Decay

A bacteria population doubles every 5 days.

a. When will it be 7.325 times as large

$$A(t) = A_i B^{\frac{t}{t_{growth}}}$$

$$(7.325)\cancel{A_i} = \cancel{A_i}(2)^{\frac{t}{5days}}$$ ← left side is multiplied by 7.325 because A_i will be 7.325 times as large.

divide A_i from both sides

$$7.325 = (2)^{\frac{t}{5days}}$$

$$\log(7.325) = \log 2^{\frac{t}{5days}}$$ ← add log to both sides

$$\log(7.325) = \frac{t}{5}\log 2$$ ← move exponent behind $\log 2$

$$\frac{5 \cdot \log(7.325)}{\log 2} = t$$ ← divide both sides by $\log 2$ multiply both sides by 5

$$14.364 = t$$

A radioactive element is decaying at the rate of 5.5% per hour. If you start with 10.810g, how much time must elapse for you to remain with 7.250g?

$$A(t) = A_i B^{\frac{t}{t_{decay}}}$$

$$7.250g = 10.810(.945)^{\frac{t}{1hr}}$$

$B = 1.00 - 0.055$

$B = 0.945$

$$\frac{7.250g}{10.81g} = 0.945^t$$

$$\log\left(\frac{7.250g}{10.81}\right) = \log(0.945)^t$$ ← add log to both sides

$$\log\left(\frac{7.250}{10.81}\right) = t\log(0.945)$$ ← move exponent behind log expression according to law of logs

$$\frac{\log\left(\frac{7.250}{10.81}\right)}{\log(0.945)} = t$$

$$7.061 = t$$

pH Solution Questions

EX 2. If an aqueous solution has a pH of 9.41, what is its concentration in moles / L?

$$pH = -\log[H^+]$$

$$-\log[H^+] = pH$$

$$\log[H^+] = -pH$$

$$10^{-pH} = [H^+]$$

$$10^{-9.41} = [H^+]$$

$$3.89 \times 10^{-10} \doteq [H^+]$$

The concentration is approximately $3.89 \times 10^{-10} \frac{mol}{L}$.

EX 3. How many times more acidic is red wine vinegar (pH 3) as compared to distilled water (pH 7) ?

$$\frac{10^{pH\ water}}{10^{pH\ vinegar}} = \frac{10^7}{10^3}$$

$$= 10^4$$

$$= 10\ 000$$

The vinegar is 10 000 times more acidic than the water.

Earthquake – Richter Scale Questions

EX 1. An earthquake is 1 000 times more intense as the reference standard. What is its magnitude on the Richter scale?

$$R = \log\left(\frac{I}{I_0}\right)$$

$$R = \log\left(\frac{1000\,\cancel{I_0}}{\cancel{I_0}}\right)$$

$$R = \log 1000$$

$$R = 3$$

The Richter value is 3.

EX 2. Chilean Quake (1960) - 9.5 on the Richter scale
Val de Bois, Quebec (2010) - 5.0 on the Richter scale

How much more intense was the Chilean earthquake compared to the one in Quebec?

$$\frac{I_C}{I_Q} = \frac{10^{R_C}}{10^{R_Q}}$$

$$= \frac{10^{9.5}}{10^{5}}$$

$$= 10^{4.5}$$

$$\doteq 31\,623$$

The Chilean quake was approximately 31 623 times more intense than the Quebec quake.

Sound – deciBel Scale Questions

EX 1. A sound is 2 300 times louder than the threshold of hearing (TOH).
How loud is it in deciBels?

$$L = 10 \log\left(\frac{I}{I_0}\right)$$

$$L = 10 \log\left(\frac{2\,300 \cancel{I_0}}{\cancel{I_0}}\right)$$

$$L = 10 \log 2\,300$$

$$L \doteq 33.6$$

The sound is approximately 33.6 dB loud.

EX 2. Whisper ..... 20 dB, Vacuum cleaner ..... 80 dB

How many times louder is the sound of a vacuum cleaner than a whisper?

$$\frac{I_{vc}}{I_w} = \frac{10^{\frac{dB\ vc}{10}}}{10^{\frac{dB\ w}{10}}}$$

$$= \frac{10^{\frac{80}{10}}}{10^{\frac{20}{10}}}$$

$$= \frac{10^8}{10^2}$$

$$= 10^6$$

$$= 1\ 000\ 000$$

The vacuum cleaner is 1 000 000 times louder than the whisper.

No long division , No angular velocity , No oblique asymptotes , Only sine / cosine functions for graphing

The exam will be in chronological order.

Short answers - 1 mark each - only the answer will be marked. Long answers - 2 to 5 marks each - must show work

Chapter	Short Answer - Marks	Long Answer - Marks
1	2	13
2	2	13
3	2	13
4	2	13
5	2	13
6	2	13
7	2	13
8	2	13
Totals	16	104
Overall	/120	